# Repair the reasoning, not just the answer

**GUIDED NOTES / use with the projected diagram** Name _____

## A SAY THE WORD

### Coordinate sign / ES: signo de la coordenada / PT: sinal da coordenada

A minus sign is not "slow." Compare sizes for speed.

Also on your note page: Distance vs. displacement, Net force, Interaction pair.

## B READ THE RELATIONSHIP

### A sign shows direction on the axis you chose. It can change the meaning of an answer.

Find the first wrong step, repair it, then recheck

## C TRACE THE ORIGINAL FIGURE

| | |
|---|---|
| 1 **CLAIM** ✗ "3 m right then 3 m back – the distance is zero." | 1 **CLAIM** ✗ "8 N pull on a 2 kg cart: 8 ÷ 2 = 4 m/s²." |
| 2 **FIX** ✓ Distance = 3 + 3 = 6 m. *Displacement* is 0 m: the endpoints match. | 2 **FIX** ✓ Net force first: 8 – 2 = 6 N. Then 6 ÷ 2 = 3 m/s². |
| 1 **CLAIM** ✗ "–2 → –6 m/s is slowing, because it's negative." | 1 **CLAIM** ✗ "The force pair cancels, so nothing moves." |
| 2 **FIX** ✓ Speed grows from 2 to 6 m/s: speeding up. The minus sign is direction. | 2 **FIX** ✓ Each pair force acts on a different cart. Draw one on each diagram. |
| ☐ **1 · PENCIL** Mistaken claims: cross out | ☐ **2 · CODE COLOR** The fixes: color by quantity |

Use the numbered key. Trace one feature. Tell your partner what it means.

## D FOLLOW THE WORKED EXAMPLE

### A 3 kg cart has 9 N right and 3 N left. A student writes 9 ÷ 3 = 3 m/s². Correct it.

**Given** m = 3 kg 9 N right 3 N left

**Rule** a = F net ÷ m

**Work** F net = 9 – 3 = 6 N right a = 6 ÷ 3

**Answer** 2 m/s² right missed: net force first

## E SAVE FOR THE FINAL MEETING

A forecast is only as good as its reasoning. Before the desk signs off on XJ, every step is checked: signs, starting position, units, and the right model.

# Read the physics: Coordinate sign

SUPPORTED READING / read one part, then point to the evidence Name _____

## 01 PICTURE IT

Write the positive direction first. Find the first step where the model, sign or starting value went wrong. Changing only the final number can hide the mistake.

## 02 HOW IT WORKS

When forces oppose, give them opposite signs before adding. When reading velocity, compare its size to tell whether speed grows. A more negative velocity can mean a larger speed.

## 03 WHAT TO CHECK

Repair the first wrong step. Then use a sketch, force sum or unit check to test the new reasoning.

## 04 TRY THE EXAMPLE FROM YOUR NOTES

A 3 kg cart has 9 N right and 3 N left. A student writes $9 \div 3 = 3$ m/s². Correct it. 2 m/s² right missed: net force first.

## 05 CONNECT TO XJ

A forecast is only as good as its reasoning. Before the desk signs off on XJ, every step is checked: signs, starting position, units, and the right model.

### SAY IT BEFORE YOU WRITE

What failed first: the rule, the sign or the arithmetic? The first error was ___. I replaced it with ___ because ___.

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Distance vs. displacement

ES: distancia y desplazamiento / PT: distância e deslocamento

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain the difference between distance and displacement?<br>In my words, this means ___. | Which quantity can be zero after a walk that covers six metres?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| Walk 4 m east and 1 m west. What distance and displacement would you record?<br>My example shows ___. | Someone reports 5 m east as the displacement. Which part of the return trip did they miss?<br>I would change ___ because ___. |

## Net force

ES: fuerza neta / PT: força resultante

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain net force to a friend?<br>In my words, this means ___. | Two people push one cart in opposite directions. Why do the arrow directions matter?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| One person pushes 6 N right and another 4 N left. What single arrow replaces both?<br>My example shows ___. | A student adds those pushes to get 10 N right. Which push did they count the wrong way?<br>I would change ___ because ___. |

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Coordinate sign

ES: signo de la coordenada / PT: sinal da coordenada

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain coordinate sign to a friend?<br>In my words, this means ___. | If right is positive, what sign goes on a 3 N force left?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| How would you label a leftward velocity of 2 m/s when right is positive?<br>My example shows ___. | Does a negative velocity by itself mean the cart is slowing down?<br>I would change ___ because ___. |

## Interaction pair

ES: par de interacción / PT: par de interação

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain interaction pair to a friend?<br>In my words, this means ___. | Earth pulls an apple down. What is the partner force, and which object feels it?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| How would you label the pair when a hand pushes a cart?<br>My example shows ___. | Are the table's push and Earth's pull on one book a third-law pair?<br>I would change ___ because ___. |

# Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

## Draw a force sum that includes a leftward force. Circle the first place where losing the sign would change your answer.

Labels you can use: first error · repair · check.

First wrong step

Repair

**Tell a partner**

The first error was ___. I replaced it with ___ because ___.

# On assignment: Coordinate sign

SECTION E / BACK / FINAL MEETING ONLY Name _____

**A calculator overlay for a 3 kg tracker shows +9 N and −3 N. Its output is "9 ÷ 3 = 3 m/s².” It is about to go live.**

## PICK A JOB / DELIVER ONE PRODUCT

☐ **Software tester**

File a bug ticket: mark the first wrong step, enter the corrected force sum and give the expected display.

☐ **Fact-check producer**

Stop or clear the graphic. Give the anchor a corrected number and a short reason viewers can check on the force arrows.

**Word bank: first error · repair · check.**

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

## INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

**“We first said ___. Now we ___ because ___.”**